COMSATS University Islamabad

COURSE DESCRIPTION FILE

EEE240 Fundamentals of Digital Logic Design

DEPARTMENT OF ELECTRICAL ENGINEERING

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Prepared By:	Checked By:	Approved By:
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Fundamentals of Digital Logic Design

Course code:

EEE240 (2+1)

Prerequisites:

None

Co requisites:

None

Course Catalog Description:

Introduction to Digital Computer and Systems, Number Systems, Binary Arithmetic, Boolean Algebra, Algebraic Manipulation, Canonical and Standard Form & Conversions, Logical Operations and Gates, Simplification of Functions, Karnaugh Map Methods, Two Level Implementations, Multi-Level NAND/NOR Circuits, Don't Care Conditions, Prime Implicants, Combinational Logic Design, Arithmetic Operations and Circuits, Analysis Procedures, Decoders, Encoders, Multiplexers, Demultiplexers, Sequential Logic, Flip-Flops, Clocked Sequential Circuits, State Machine Concept, Design of Sequential Circuits using State Machines.

Textbooks:

1. M. Morris Mano and Michael D. Ciletti, Digital Design: With an Introduction to Verilog HDL, Fifth Edition, Prentice Hall, 2012.

Reference Books:

1. Stephen Brown and Zvonko Vranesic, Fundamentals of Digital Logic with Verilog Design, Third Edition, McGraw-Hill, 2013.

Course Learning Objectives:

This is a basic course which focuses on the basic techniques of digital circuit designing. The students will learn different techniques to design simple to moderate level circuits. The course contains extensive lab work, in which students will learn to design at gate level logic circuits and simulate it on Electronic Design Automation (EDA) software tools such as Proteus, CircuitLogix, CircuitLab, LTspice etc.

Course Learning Outcomes:

After successfully completing the course, the students will be able to:

1. Demonstrate the understanding of number systems, binary arithmetic operations, codes and working principle of logic gates. **(C2-PLO1)**
2. Apply Boolean Algebra and K-Map techniques for gate level minimization of digital circuits, and represent the digital functions in canonical and standard forms. **(C3-PLO1)**
3. Analyze combinational and synchronous sequential logic circuits. **(C4-PLO2)**
4. Design combinational and synchronous sequential circuits. **(C5-PLO3)**

5. Use modern EDA tools for simulation of digital circuits and trainers/ICs for their implementation. **(P3-PLO5)**

Course Schedule:

2 credit hours/week
One laboratory session/week (3 hours/session)

Topics Covered:

1. Fundamental digital concepts, Boolean algebra
2. Number Systems and codes, synthesis using logic gates
3. Fundamentals of NAND-NOR Implementations
4. Optimized Implementation of logic functions using K-Maps
5. Analysis of combinational circuits
6. Design procedure of combinational circuits
7. Sequential circuits and flip-Flops
8. Sequential circuit design and state machines
9. Synthesis using D-flip flops, J/K and T flip-flops
10. Analysis of sequential circuits

Assessment Plan:

Theory	Quizzes (at least 4)		15%
	Homework assignments (at least 4)		10%
	Mid-term exam (in class, 60-80 minutes)		25%
	Terminal exam (3 hours)		50%
		Total (theory)	**100%**
Lab work	Lab reports (12)		25%
	Lab Mid-term exam		25%
	Project/Terminal Exam		50%
		Total (lab)	**100%**
Final marks	Theory marks * 0.67 + Lab marks * 0.33		

Learning Outcomes Assessment Plan (Tentative):

Sr. #	Course Learning Outcomes	Assessment
1.		Quiz 1
2.		Quiz 2
3.		Quiz 3
4.		Quiz 4
5.		Assignment 1

6.		Assignment 2
7.		Assignment 3
8.		Assignment 4
9.		Mid-term Exam
10.		Terminal Exam

Table 1: Assessment Plan for Course Learning Outcomes

Laboratory Experiences:

There is a Laboratory component in all 2+1 credit courses taught at the department. Lab work consists of a minimum of 12 experiments and related assignments, which constitute 25% of the overall course-grade. The laboratory experiments include implementation of combinational and sequential circuits taught in class using EDA tools such as Proteus and trainers/ICs for their implementation.

Laboratory Resources:

The relevant laboratory is equipped with workbenches and computers to facilitate the experiments outlined in the lab handbook(s) that are periodically updated. A current list of the 12 lab experiments performed in this course is provided as Annexure-II. The list of software and equipment available is also posted in all labs and is managed by staff dedicated for this purpose.

Computer Resources:

The laboratory systems are equipped with digital design and simulation software for facilitating the simulation of digital circuits for this course.

Mapping Course Learning Outcomes (CLOs) to Program Learning Outcomes (PLOs):

PLO 1	**Engineering Knowledge:** An ability to apply knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.
PLO 2	**Problem Analysis:** An ability to identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.
PLO 3	**Design/Development of Solutions:** An ability to design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.
PLO 4	**Investigation:** An ability to investigate complex engineering problems in a methodical way including literature survey, design and conduct of experiments, analysis and interpretation of experimental data, and synthesis of information to derive valid conclusions.
PLO 5	**Modern Tool Usage:** An ability to create, select and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modeling, to complex engineering activities, with an understanding of the limitations.
PLO 6	**The Engineer and Society:** An ability to apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solution to complex engineering problems.
PLO 7	**Environment and Sustainability:** An ability to understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate knowledge of and need for sustainable development.
PLO 8	**Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.
PLO 9	**Individual and Team Work:** An ability to work effectively, as an individual or in a team, on multifaceted and /or multidisciplinary settings.
PLO 10	**Communication:** An ability to communicate effectively, orally as well as in writing, on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PLO 11	**Project Management:** An ability to demonstrate management skills and apply engineering principles to one’s own work, as a member and/or leader in a team, to manage projects in a multidisciplinary environment.
PLO 12	**Lifelong Learning:** An ability to recognize importance of, and pursue lifelong learning in the broader context of innovation and technological developments.

PLOs / CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO 9	PLO10	PLO11	PL012
CLO1	C2											
CLO2	C3											
CLO3		C4										
CLO4			C5									
CLO5					P3							

Table 2: Mapping CLOs to PLOs

PLO Coverage Explanation:

PLO 1: Engineering Knowledge
The homework, exams, and laboratory experiments require direct application of mathematics, scientific, and engineering knowledge to successfully complete the course. This includes function minimization using Boolean algebra, K-Maps and state tables. (High relevance to course)

PLO 2: Problem Analysis
The final design is given as a set of specifications that the students' design must meet. Therefore, they must identify the key limiting issues, formulate a solution strategy, research and test their approach, and finally prototype and analyze the design to prove that it works. (High relevance to course)

PLO 3: Design/Development of Solutions
Through design and implementation of combinational and synchronous sequential circuits the students enhance their design and development ability. (High relevance to course)

PLO 5: Modern Tool Usage
The Proteus Simulation tool is introduced and used in the lab to simulate different combinational and sequential circuits. (High relevance)

PLO 4, 6 - 12 are not covered and checked in the course directly. However, certain aspects are covered directly or indirectly. For example project management skill is developed during planning of different stages of the design projects. Also, various written and oral communication skills are also checked in the lab reports. (Low relevance to course)

ANNEXURE-I

Tentative Lecture Breakdown (30 Lectures):

Topics	No. of Lectures
Introduction to Digital Systems, binary numbers, and Number base conversions, Binary arithmetic operations, Complements of numbers, and subtraction with compliments, Binary Logic, Theorems and Postulates of Boolean Algebra, Operator Precedence.	4
Digital Logic Gates, Logic Levels, Boolean functions, (NAND-NOR Implementation), Algebraic Manipulation, Function Complement, Canonical and Standard Forms, Signed Binary Numbers.	4
K-Maps, Prime Implicants, Essential Prime Implicants, Product of Sum Simplification and Don't care Conditions.	3
Combinational Logic: Analysis Procedure and Design Procedure, Binary Adder/Subtractor, BCD Adder, Binary Multiplier, Magnitude Comparator and Decoders, Encoders, Parity Encoders, Mux, Boolean Function Implementation using MUX.	9
Sequential Circuits, SR Latch (NAND/NOR), D Latch, Types and characteristics of Flip Flops (D, J-K, T), Analysis of clocked Sequential circuits (State Equations, State Table, State Diagram).	5
Mealy and Moore models of Finite State Machines, State reductions and State Assignments, Design Procedure of Sequential Circuits, Synthesis using D-F/F (Sequence Detector Non-Overlapping), Excitation Tables, Pattern/Sequence Detector (Overlapping), Synthesis using J/K, T F/F.	5

ANNEXURE-II

List of Experiments:

Lab No.	Details
1	Introduction to Basic Logic Gate and Proteus Simulation
2	Boolean Function Implementation using Universal Gates
3	Design and Implementation of Multi-Level NAND/NOR Circuits
4	Design and Implementation of Boolean Functions By Standard Forms
5	Logic Minimization of Complex Functions using Automated Tools
6	Design and Implementation of an $n - bit$ Adder/Subtractor
7	Design and Implementation of an $n - bit$ Multiplier
8	Design and Implementation of a Boolean function using Multiplexer
9	Design and Implementation of Binary to BCD to 7-Segment Decoder on Proteus
10	Design and Implementation of a non-overlapping Sequence Detector using Mealy/Moore Machine (D Flip/Flop)
11	Design and Implementation of a non-overlapping Sequence Detector using Mealy/Moore Machine (JK and T Flip/Flop)
12	Design and Implementation of a overlapping Sequence Detector using Mealy/Moore Machine
13-14	Lab Project / Viva

Version	Applicable From
Version 1	Spring 2024